

# Quantifying Exposome-Health Associations with Bayesian Multiple Index Models

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# BMIM: A Novel Approach for Mixtures Analyses

Our approach combines the strengths of two popular mixtures methods:

1. Linear index models

- E.g. Weighted quantile sum regression (WQS), Quantile g-Computation (QGC)

2. Response surface methods

- E.g. Bayesian kernel machine regression (BKMR)

# Linear Index Methods

Linear on the quantile scale

$$y_i = \alpha_0 + \beta \times [b_1 q_{i1} + \cdots + b_P q_{iP}] + \mathbf{z}_i^T \boldsymbol{\gamma} + \epsilon_i$$

- i.e.  $q_{i1} = 0, \dots, 9$  for deciles

Decompose coefficients of a linear model into

- $\beta$ : an “overall mixture effect”
- $b_p$ : component weights

Very interpretable

- Weights  $b_p$  indicate relative contribution of each component

Somewhat restrictive:

- Assume linearity (on the quantile scale)
- Assume no interactions between any exposures

Different versions

- Constrained: Weighted quantile sum regression (Carrico et al., 2015)
- Unconstrained: Quantile g-computation (Keil et al., 2020)

# Bayesian Kernel Machine Regression (BKMR)

Given  $P$  exposures,  $x_1, \dots, x_P$ ,

$$y_i = h(x_{i1}, \dots, x_{iP}) + \mathbf{z}_i^T \boldsymbol{\gamma} + \epsilon_i,$$

$h(\cdot)$ : is some  $P$ -dimensional unknown exposure-response function

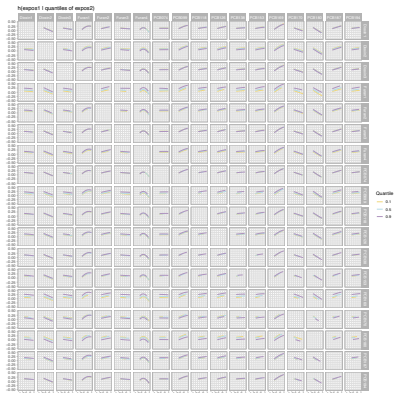
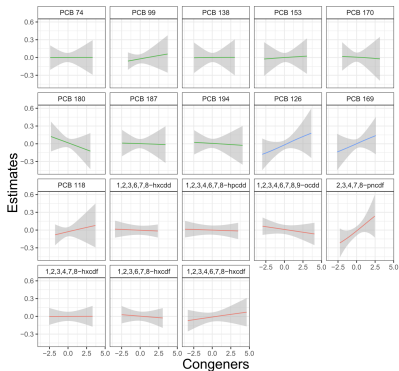
- Allows non-linearity exposure-response curves
- Allows high-order interactions among exposures

Interpret results by estimating exposure-response curves

- Plot  $P$  different exposure-response curves
- Consider  $P \times (P - 1)/2$  two-way interactions

Bobb et al (2015,2018)

# Example: NHANES ( $P = 18$ )



Gibson et al. (2019)

# A Tradeoff

Linear index methods:

- Easy to interpret
- Somewhat restrictive
  - Assumes linearity and no interactions

BKMR:

- Very flexible
  - Allows non-linearities and higher order interactions
- Difficult to interpret when there are many exposures ( $P$  is large)

# Proposed Bayesian Multiple Index Models (BMIM)

Partition  $x_{i1}, \dots, x_{iP}$  into  $M$  groups  $\mathbf{x}_{im} = (x_{im1}, \dots, x_{imL_m})^T$ .

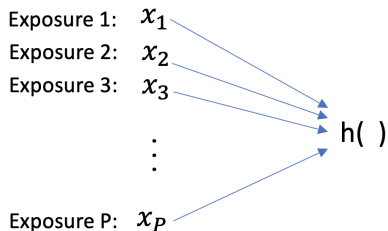
$$\text{The BMIM is: } y_i = h\left(\sum_{l=1}^{L_1} x_{i1l}\theta_{1l}, \dots, \sum_{l=1}^{L_M} x_{iMl}\theta_{Ml}\right) + \mathbf{z}_i^T \boldsymbol{\gamma} + \epsilon_i$$

$h(\cdot)$  is a function of  $M$  **indices**  $E_m = \sum_{l=1}^{L_m} x_{iml}\theta_{ml}$

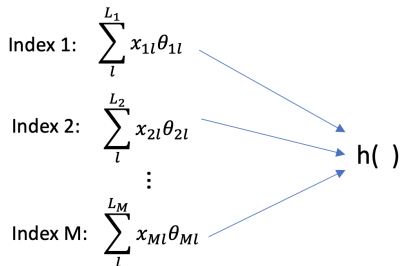
- Allows non-linearity of **indices**
  - Response curves of exposures within an index have the same shape
- Allows interactions among **indices**
  - Assumes no interactions within indices

# Reducing Dimensionality

**BKMR:** dimension  $P$



**BMIM:** dimension  $M \leq P$



# Interpreting the BMIM

Interpret results by estimating indexwise-response curves

- Plot  $M \leq P$  different indexwise response curves
- Consider two-way interactions

Estimate index weights  $\theta_{ml}$

- Quantify relative contributions of each  $l^{th}$  component of the  $m^{th}$  index
- Subject to identifiability constraints:  $\sum_{l=1}^{L_m} \theta_{ml}^2 = 1$  and  $\sum_{l=1}^{L_m} \theta_{ml} \geq 0$

# Exposome Analysis: Applying the BMIM

**Outcome** of Interest: BMI z-score

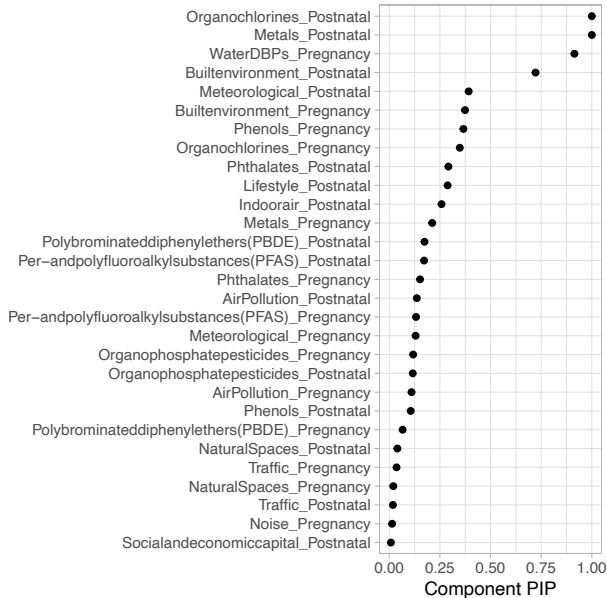
150 continuous **exposures**:

- Grouped into **29 indices** corresponding to exposure domains/families
  - E.g. organochlorines, metals, phthalates etc.
  - Distinct indices for pregnancy and postnatal exposures
- Subject to Bayesian variable selection

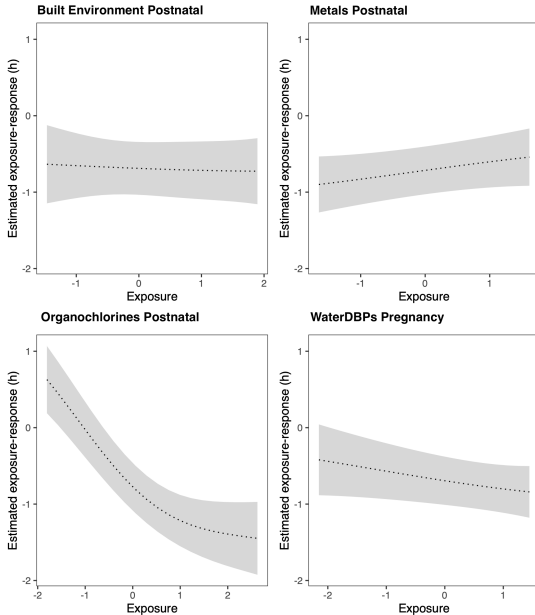
Adjusted for **covariates**:

- Cohort, gestational age at birth, maternal BMI, maternal weight gain, maternal education, parity, native to country of birth

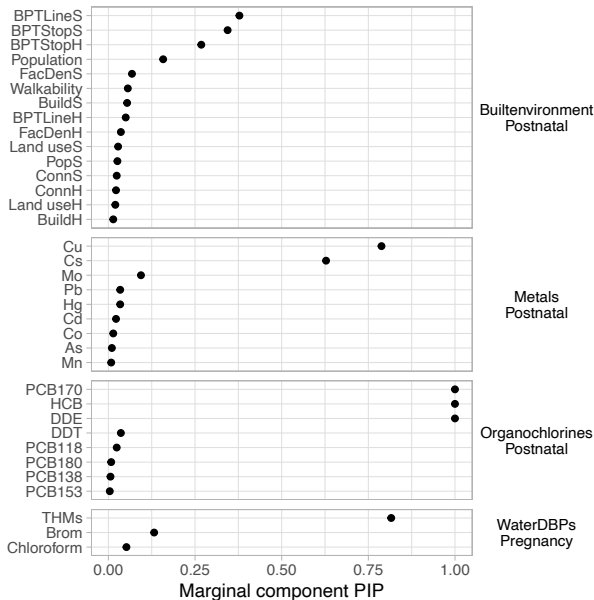
# Exposome Analysis: Indexwise PIPs



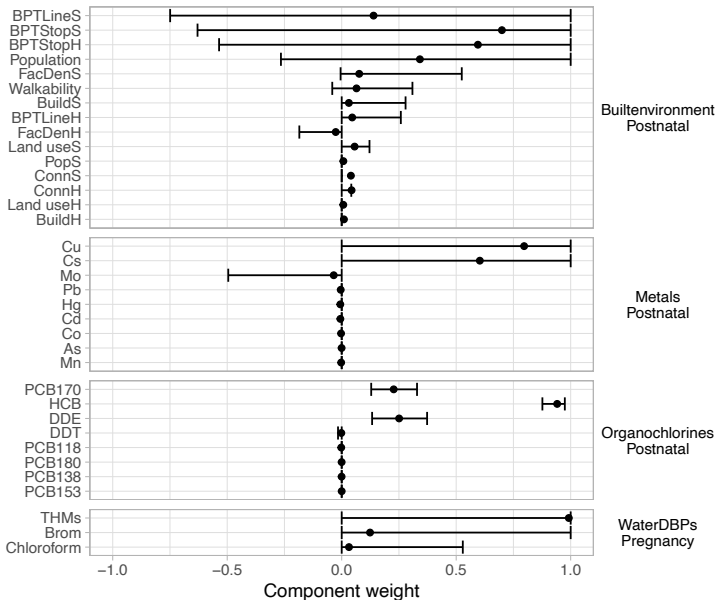
# Exposome Analysis: Indexwise Response Curves



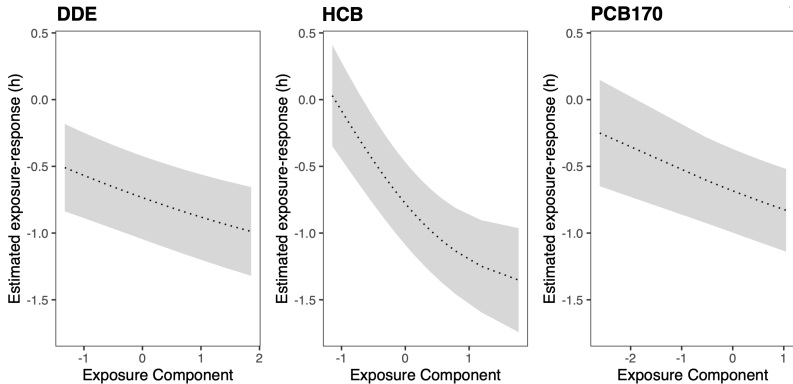
# Exposome Analysis: Componentwise PIPs



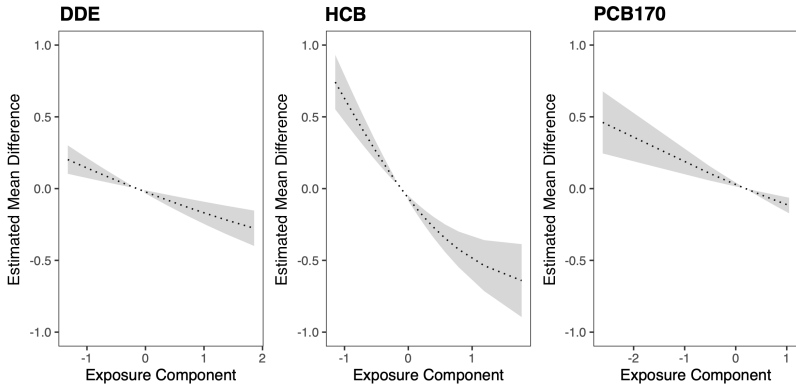
# Exposome Analysis: Index Weights



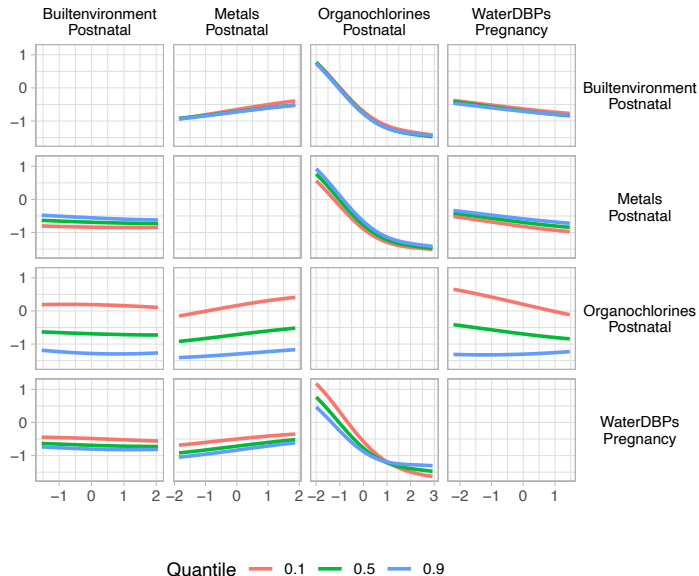
# Exposome Analysis: Componentwise Response Curves



# Exposome Analysis: Componentwise Contrasts



# Exposome Analysis: Indexwise Interactions



# Limitations

## Assumptions

- Response curves for exposures within same index take same shape
- No (epidemiological) interaction among exposures within the same index

Variable selection may conceal small but non-null effects

## Computing time

- MCMC requires inverting  $N \times N$  matrices

# Exposome Analysis: BMIM Summary & Extensions

BMIM represents a spectrum of models varying in flexibility and interpretability

- SIM and BKMR as special cases
- Allows for other useful structures

E.g. interactions between multipollutant index and individual covariates

Extensions:

- Constraints on index weights
- DLM structure for time-varying exposures (Wilson et al., 2019)

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McGee, G., Wilson, A., Webster, T. F., & Coull, B. A. (2021). Bayesian Multiple Index Models for Environmental Mixtures. arXiv preprint:2101.05352.

[github.com/glenmcgee/bsmim2](https://github.com/glenmcgee/bsmim2)

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